

Project Initialization and Planning Phase

Date	
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to predict the Human Development Index (HDI) of a country using Machine Learning techniques. The system analyzes important development indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita to estimate the HDI score. The application provides accurate predictions, classifies countries into HDI categories, and supports real-time prediction through a user-friendly Flask web application.

Project Overview	
Objective	The primary objective of this project is to develop a machine learning-based web application that accurately predicts the Human Development Index (HDI) using important socio-economic indicators, helping users understand a country's level of human development.
Scope	The project includes data preprocessing, model training using Linear Regression, model deployment through Flask, real-time prediction, input validation, and HDI category classification using historical HDI datasets.
Problem Statement	
Description	Existing methods for calculating and analyzing HDI require extensive statistical analysis and manual processing. There is a need for an intelligent automated system that predicts HDI accurately using machine learning techniques.
Impact	The proposed system reduces manual effort, improves prediction accuracy, enables faster analysis, and supports decision-making for researchers, students, and policymakers.
Proposed Solution	
Approach	The system uses a trained Linear Regression model to predict HDI scores based on user inputs such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The Flask application validates the inputs, performs prediction, and displays the HDI score along with its category.

Key Features	• Machine Learning-based HDI Prediction • Flask Web Application • Country-wise Auto-fill Feature • User Input Validation • Real-time Prediction • HDI Category Classification • Fast Prediction using Pickle Model • User-friendly Interface
	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5
Memory	RAM	8 GB
Storage	Disk space	256 GB SSD or above
Software		
Frameworks	Web frameworks	Flask
Libraries	ML library	Scikit-learn
Development Environment	IDE	Jupyter Notebook
Data		
Data	Source, size, format	Cleaned HDI Dataset (CSV Format) Kaggle / UNDP Human Development Dataset CSV